

Caleb Erickson

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Education

Master of Business Administration, Cybersecurity Focus May 2027

The University of Alabama, Manderson Graduate School of Business, Tuscaloosa, AL

- GPA: 4.0/4.0
- STEM Path to MBA

Bachelor of Science in Computer Science May 2026

The University of Alabama, Lee J. Styslinger Jr. College of Engineering, Tuscaloosa, AL

- GPA: 4.0/4.0
- Summa Cum Laude
- President's List
- Honors College
- Most Outstanding Undergraduate Award

Skills

- **Languages:** Python, C#, C++, C, TypeScript, JavaScript
- **Frameworks:** FastAPI, Django, Flask, .NET, React, Next.js, Node.js
- **Cloud and Tools:** AWS Lambda, Amazon RDS, CloudFront, Azure, Google Cloud, Docker, Proxmox, Git, Jira
- **Databases:** PostgreSQL, MongoDB, Azure Cosmos DB, Neo4j, Milvus
- **Data and ML:** NumPy, pandas, Matplotlib, scikit-learn

Experience

CGI - Software Developer Intern June 2026 – July 2026

- Collaborated with a 10-person intern team to develop an internal promotion-tracking tool used by 2,000 CGI partners
- Served as Scrum Master for Sprint 2, increasing backlog completion from 56% to 90% despite an 80% larger backlog than Sprint 1 by clarifying requirements, removing duplicate work, and standardizing development practices
- Implemented least-privilege role-based access control for three user types using internal API data, frontend route protection, and backend authorization middleware

Alabama AI Institute - AI Researcher April 2025 – Present

- Built an automated Azure-based ETL pipeline integrating 5 federal datasets into a 50,000-node knowledge graph
- Developed a RAG framework combining vector embeddings, language models, and knowledge graphs, reducing hallucinations by 38% through reranking, optimized document chunking, and improved embedding quality
- Led development of an AI-assisted data storytelling app used by ~75 users across six university courses
- Presented and demonstrated AI systems to ~250 attendees at the NSF OKN Year 3 Kickoff in Washington, D.C.

Projects

Mockstreet Stock Exchange, *mockstreetexchange.com* March 2025

- Built a simulated stock exchange that enables users to simulate trades and manage portfolios using real-time data
- Architected an AWS deployment using Lambda, Amazon RDS, and CloudFront to support 100 concurrent users

Number Prediction Neural Network, *numbers-beige.vercel.app* April 2025

- Designed and trained a convolutional neural network from scratch using NumPy, achieving 96.4% accuracy on the MNIST dataset, and developed a browser-based interface for real-time handwritten-digit predictions

Portfolio with All Projects, *caleberickson21.github.io*

Publications

• Co-author, "Understanding Student Engagement and Achievement for AI-Mediate Data Storytelling Workflows," *ACM Conference on International Computing Education Research (ICER)* August 2026

• Co-author, "Operationalizing Generative Spatial Intelligence: A Conversational GeoAI Framework for Flood Risk Communication," *Research Square (preprint)* July 2026

• Co-author, "A Semi-Automated Framework for Flood Ontology Construction with an Application in Risk Communication," *Water* September 2025